

hw4

June 29, 2024

Load the data in 2023 and create a SQLite database.

```
[23]: import pandas as pd
import sqlite3
import os

csv_file = 'D:/472/hw4/data/2023.csv'
table_name = 'Data2023'

db_file = 'data2023.db'
conn = sqlite3.connect(db_file)

df = pd.read_csv(csv_file, usecols=[0, 1, 2, 3])
df.columns = ['id', 'day', 'type', 'data']
df.to_sql(table_name, conn, if_exists='replace', index=False)

conn.close()
```

```
[23]: txt_file = 'D:/472/hw4/data/ghcnd-stations.txt'
table_name = 'Stations'

db_file = 'data2023.db'
conn = sqlite3.connect(db_file)

data = []
with open(txt_file, 'r') as file:
    for line in file:
        s_id = line[0:11].strip()
        s_name = line[41:71].strip()
        data.append([s_id, s_name])

df = pd.DataFrame(data, columns=['s_id', 's_name'])
df.to_sql(table_name, conn, if_exists='replace', index=False)

conn.close()
```

```
[1]: %matplotlib inline
%load_ext sql
```

```
[2]: %%sql
sqlite:///data2023.db
```

Connecting to 'sqlite:///data2023.db'

```
[3]: %%sql
SELECT * FROM sqlite_master WHERE type='table';
```

Running query in 'sqlite:///data2023.db'

```
[3]: +-----+-----+-----+-----+-----+
| type | name | tbl_name | rootpage | sql |
+-----+-----+-----+-----+-----+
| table | Data2023 | Data2023 | 2 | CREATE TABLE "Data2023" ( |
| | | | | "id" TEXT, |
| | | | | "day" INTEGER, |
| | | | | "type" TEXT, |
| | | | | "data" INTEGER |
| | | | | ) |
| table | Stations | Stations | 293143 | CREATE TABLE "Stations" ( |
| | | | | "s_id" TEXT, |
| | | | | "s_name" TEXT |
| | | | | ) |
+-----+-----+-----+-----+-----+
```

What are the top 3 highest recorded temperatures in 2023?

```
[22]: %%sql query_q1 <<
SELECT Stations.s_name, Data2023.day, Data2023.data / 10.0 AS adjusted_data
FROM Data2023
JOIN Stations
ON Data2023.id = Stations.s_id
WHERE Data2023.type = 'TMAX' AND Data2023.data < 1000
ORDER BY Data2023.data DESC
LIMIT 3;
```

Running query in 'sqlite:///data2023.db'

```
[23]: query_q1.DataFrame()
```

```
[23]:      s_name      day  adjusted_data
0  WHITE WOLF CALIFORNIA  20230611         65.6
1  WHITE WOLF CALIFORNIA  20230610         63.9
2  PARK RIDGE CALIFORNIA  20230608         61.7
```

The hottest temperature in 2023 was recorded at the station 'WHITE WOLF CALIFORNIA' on 2023-06-11, with a temperature of 65.6 degrees Celsius.

What are the top 3 lowest recorded temperatures in 2023?

```
[16]: %%sql query_q2 <<
SELECT Stations.s_name, Data2023.day, Data2023.data / 10.0 AS adjusted_data
FROM Data2023
JOIN Stations
ON Data2023.id = Stations.s_id
WHERE Data2023.type = 'TMIN' AND Data2023.data > -9000
ORDER BY Data2023.data ASC
LIMIT 3;
```

Running query in 'sqlite:///data2023.db'

```
[17]: query_q2.DataFrame()
```

```
[17]:      s_name      day  adjusted_data
0  VOSTOK  20230807          -82.0
1  VOSTOK  20230812          -81.0
2  VOSTOK  20230909          -80.6
```

The coldest temperature in 2023 was recorded at the station 'Vostok' (the Russian research station in the interior of Antarctica) on 2023-08-07, with a temperature of -82.0 degrees Fahrenheit.

Find the top 3 temperatures in Shanghai in 2023.

```
[20]: %%sql query_q3 <<
SELECT Stations.s_name, Data2023.day, Data2023.data / 10.0 AS adjusted_data
FROM Data2023
JOIN Stations
ON Data2023.id = Stations.s_id
WHERE Stations.s_name = 'SHANGHAI' AND Data2023.type = 'TMAX' AND Data2023.data_
↪ < 2000
ORDER BY Data2023.data DESC
LIMIT 3;
```

Running query in 'sqlite:///data2023.db'

```
[21]: query_q3.DataFrame()
```

```
[21]:      s_name      day  adjusted_data
0  SHANGHAI  20230712          38.7
1  SHANGHAI  20230713          38.7
2  SHANGHAI  20230711          38.0
```

The highest temperature recorded at the station 'SHANGHAI' in 2023 was 38.7 degrees Celsius.